Supplementary Materials to the manuscript entitled:

Carbon on the nanoscale: ultrastiffness and unambiguous definition of incompressibility

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Section S1. Derivation of the relation between average bond stiffness and bulk modulus

We consider two quantities – bulk modulus and average bond stiffness – both being the second derivatives of total energy:

$$B = V\left(\frac{\partial^2 E}{\partial V^2}\right); \langle k\rangle = \frac{1}{N_b}\left(\frac{\partial^2 E}{\partial \langle l\rangle^2}\right) \quad \text{(S1)}$$

These definitions can be rewritten using linear and volumetric strains $\varepsilon = \frac{\langle l\rangle - \langle l\rangle_0}{\langle l\rangle_0}$ and $\delta = \frac{V - V_0}{V_0}$:

$$B = \frac{V}{V_0^2}\left(\frac{\partial^2 E}{\partial \delta^2}\right) = \frac{1+\delta}{V_0}\left(\frac{\partial^2 E}{\partial \delta^2}\right); \langle k\rangle = \frac{1}{N_b\langle l\rangle_0^2}\left(\frac{\partial^2 E}{\partial \varepsilon^2}\right) \quad \text{(S2)}$$

The mathematical relation between the derivatives is given by a simple formula:

$$\frac{\partial^2 E(\delta(\varepsilon))}{\partial \varepsilon^2} = \frac{\partial^2 E}{\partial \delta^2}\cdot\left(\frac{\partial \delta}{\partial \varepsilon}\right)^2 + \frac{\partial E}{\partial \delta}\cdot\left(\frac{\partial^2 \delta}{\partial \varepsilon^2}\right) \quad \text{(S3)}$$

Expressing the second derivatives from Eq. (S2) and regarding pressure as $P = -\frac{1}{V_0}\frac{\partial E}{\partial \delta}$, we finally obtain:

$$\langle k\rangle = \frac{V_0}{N_b\langle l\rangle_0^2}\left[\frac{B}{1+\delta}\left(\frac{\partial \delta}{\partial \varepsilon}\right)^2 - P\left(\frac{\partial^2 \delta}{\partial \varepsilon^2}\right)\right] \quad \text{(S4)}$$

At zero pressure, this reduces to:

$$B_0^{est} = \langle k\rangle_0 \cdot \frac{N_b\langle l\rangle_0^2}{V_0\left(\frac{\partial \delta}{\partial \varepsilon}\right)^2_{\varepsilon=0}} \quad \text{(S5)}$$

Section S2. Auxiliary figures about bulk modulus calculations

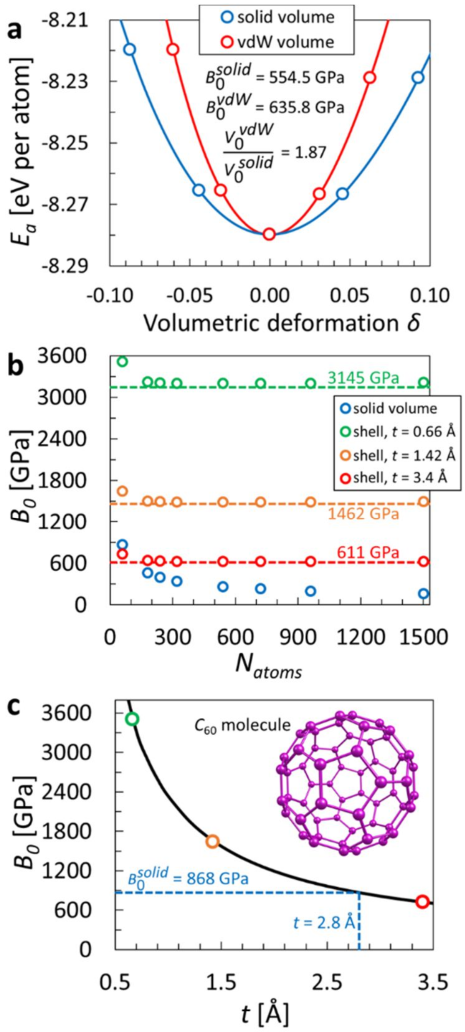

Figure S1. (a) Typical energy vs. deformation curve at the example of C_{259} nanodiamond. Data for both considered volume definitions are represented. Birch-Murnaghan fit to the data is shown with solid lines. (b) The bulk modulus of fullerenes calculated using the elastic shell approach with various shell thicknesses. Dashed lines display values of effective bulk modulus for graphene with the corresponding layer thickness. The results for the solid medium approach are also shown for comparison. As one can see, any bulk modulus in the range 0.6 – 3.5 TPa can be obtained for fullerenes depending on the chosen shell thickness. (c) Hyperbolic dependence of C_{60} bulk modulus on a shell thickness considering an elastic shell model. Prediction of B_0 = 868 GPa using solid medium volume numerically matches the result of the elastic shell model with the thickness t = 2.8 Å.

Section S3. Technical details on the calculation of $\langle k \rangle_0^{est}$

To determine the most appropriate type of hybridization for each bond in the structure, we introduced metrics which measure how the chemical bond is close to sp^2- or sp^3-hybridized bond: $M_{sp^2} = \frac{|l - l_{sp^2}|}{l_{sp^2}} + \frac{1}{m}\sum_i^m \frac{|\alpha_i - \theta_{sp^2}|}{\theta_{sp^2}} + \frac{1}{n}\sum_i^n \frac{|\beta_i - \theta_{sp^2}|}{\theta_{sp^2}}$, $M_{sp^3} = \frac{|l - l_{sp^3}|}{l_{sp^3}} + \frac{1}{m}\sum_i^m \frac{|\alpha_i - \theta_{sp^3}|}{\theta_{sp^3}} + \frac{1}{n}\sum_i^n \frac{|\beta_i - \theta_{sp^3}|}{\theta_{sp^3}}$. Here l is the length of the bond between atoms A and B, m and n are the numbers of A and B nearest neighbors, α_i and β_i are the bond angles adjacent to the bond line from the side of atom A and B, respectively. Parameters $l_{sp^2} = 1.424$ Å, $\theta_{sp^2} = 120°$ and $l_{sp^3} = 1.547$ Å, $\theta_{sp^3} = 109.47°$ correspond to bond lengths and bond angles in graphene and bulk diamond, as we found from our DFT calculations. The nearest-neighbor cutoff was set to 1.8 Å. Values of M_{sp^2} and M_{sp^3} were computed for each bond in the structure and then compared. If $M_{sp^2} > M_{sp^3}$, sp^3-hybridization was assigned to the bond, and sp^2-hybridization was determined otherwise. We also tested several various forms of these metrics (*e.g.*, with $\sin|\alpha_i - \theta_{sp^2}|$ instead of $|\alpha_i - \theta_{sp^2}|/\theta_{sp^2}$) and revealed that results are not sensitive to the specific functional form of the metrics.

Section S4. Estimates of bulk modulus for various volume definitions

The Eq. (S5) enables to recalculate average bond stiffness into bulk modulus for a given relationship $\delta(\varepsilon)$ under hydrostatic compression, that depends on the definition of volume considered. In the simplest cases of cubic crystal and solid medium volume V^{solid}, the dependence is straightforward: $\delta(\varepsilon) = 3\varepsilon + 3\varepsilon^2 + \varepsilon^3$, and Eq. (S5) leads to the following expression for bulk modulus:

$$B_0^{est} = \langle k \rangle_0 \cdot \frac{N_b \langle l \rangle_0^2}{9V_0^{solid}} \tag{S6}$$

We also considered van der Waals V^{vdW} volume for nanodiamonds to take into account the finite size of carbon atoms. For clarity and simplicity, we derive $\delta(\varepsilon)$ dependence only for nanocluster of cubic shape. Its volume V^{vdW} can be approximated as a solid medium volume of a cube with its edge a_0 elongated by the value of $2r_C$ (see the scheme in Figure 4b from the main text): $V^{vdW} \approx (a_0 + 2r_C)^3$. Subjected to linear deformation ε, cube edge length initially determined as $a_0 = (V_0^{solid})^{1/3}$ should become equal to $a = a_0(1 + \varepsilon)$. Thus, for volumetric strain and bulk modulus we have:

$$\delta = \frac{((a_0 + 2r_C) + \varepsilon a_0)^3 - (a_0 + 2r_C)^3}{(a_0 + 2r_C)^3} = 3\left(\frac{\varepsilon}{\lambda}\right) + 3\left(\frac{\varepsilon}{\lambda}\right)^2 + \left(\frac{\varepsilon}{\lambda}\right)^3, \lambda = 1 + \frac{2r_C}{a_0}$$

$$B_0^{est} = \langle k \rangle_0 \cdot \frac{N_b \langle l \rangle_0^2}{9V_0^{vdW}} \cdot \left(1 + \frac{2r_C}{a_0}\right)^2. \quad \text{(S7)}$$

Elastic shell volume V^{shell} can be easily found as a volume enclosed between two spheres of radii $R - t/2$ and $R + t/2$ (Figure 4c): $V^{shell} = 4\pi R^2 t + \frac{4\pi}{3}\left(\frac{t}{2}\right)^3 \approx 4\pi R^2 t$, assuming $t \ll 2R$. Next, linear strain $\varepsilon = \frac{R - R_0}{R_0}$ transforms to dilatation $\delta = \frac{4\pi R_0^2 t\,((1+\varepsilon)^2 - 1)}{4\pi R_0^2 t} = 2\varepsilon + \varepsilon^2$. For bulk modulus, this results in:

$$B_0^{est} = \langle k \rangle_0 \cdot \frac{N_b \langle l \rangle_0^2}{4V_0^{shell}}. \quad \text{(S8)}$$

The results in Figure 4a show that Eq. (S6) gives slightly overestimated values of bulk modulus. Nevertheless, the discrepancies between B_0^{est} and B_0 do not exceed 8%, and the error reduces with increasing size, whereas for fullerenes B_0^{est} and B_0 coincide perfectly.

Although Eq. (S7) was derived only for cubic shape, it could be made applicable also for cuboctahedral nanodiamonds by introducing their effective size as $a_0 = (V_0^{solid})^{1/3}$ in consistence with the case of cubic shape. Figure 4b demonstrates that Eq. (S7) tends to underestimate bulk modulus, and the results are a bit more scattered than the predictions of Eq. (S6) with the maximum variance of 14% between B_0^{est} and B_0. However, again the error reduces with increasing size and the suggested model gives acceptable estimates.

As for the elastic shell model, the ratio B_0^{est}/B_0 rapidly converges to unity with the enlargement of fullerenes (Figure 4c). For the smallest C_{60} molecule, the discrepancy between B_0^{est} and B_0 is 14% since values of t (3.4 Å) and $2R$ (7.1 Å) are comparable. For the remaining fullerenes, this model gives a very good estimate of bulk modulus with an accuracy within 5%.

Section S5. Limitations of average bond stiffness approach

S5.1. Structures with van der Waals bonding

Besides considered in the main text cubic and cuboctahedral shapes, octahedral and truncated octahedral nanodiamonds were also studied in previous theoretical works [1,2]. The surface of such nanodiamonds is mainly composed of {111} diamond faces (100% and 76%, respectively).

We considered three octahedral (C_{84}, C_{165}, C_{455}) and two truncated octahedral (C_{548} and C_{837}) nanodiamonds and performed all the same calculations that are described in the main text.

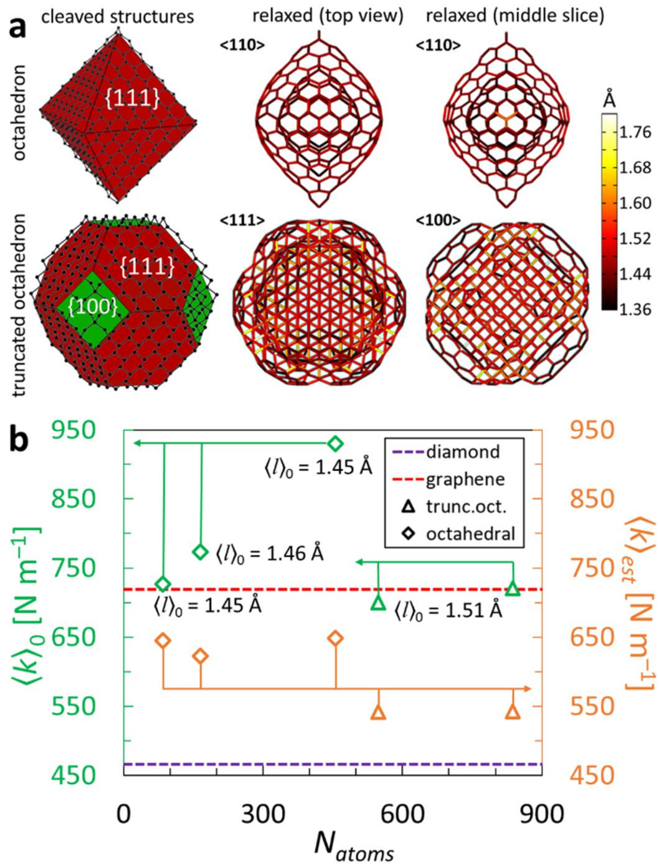

Figure S2. (a) Examples of octahedral (C_{455}) and truncated octahedral (C_{837}) shapes of nanodiamonds. Graphitization is observed due to a predominant presence of {111} diamond planes on the surface. (b) $\langle k \rangle_0$ and $\langle k \rangle_0^{est}$ calculated for nanodiamonds of these shapes. Average bond length $\langle l \rangle_0$ is given for each structure. The accordance between axes and groups of data points is shown with the arrows of the corresponding color.

As expected, after structural relaxation octahedral nanodiamonds turned into the onion-like structures with a small core containing distorted sp^3-bonds which is surrounded by several concentric cages (Figure S2a), almost entirely composed of sp^2-hybridized atoms. This effect of graphitization in diamond is a well-known and experimentally observed phenomenon [3]. With a smaller proportion of {111} surfaces effect of graphitization become less pronounced in truncated octahedral nanodiamonds. The delamination of {111} segments is still observed, although {100} areas remain covalently bonded.

We examined the mechanical properties of these structures and revealed implausibly high values of average bond stiffness (Figure S2b). Such high $\langle k \rangle_0$ exceeding or close to bond stiffness of graphene cannot be expected for them, as their average bond lengths $\langle l \rangle_0$ are all larger than the equilibrium value for graphene (1.425 Å), and therefore $\langle k \rangle_0$ of considered nanodiamonds should be lower than that of graphene. Calculations of $\langle k \rangle_0^{est}$ basing on the relaxed bond lengths distribution confirm this statement (Figure S2b). Such unphysical results obtained from direct calculations of $\langle k \rangle_0$ are all due to graphitization and delaminated atomic structure. Indeed, only surficial atoms were frozen in calculations to simulate the applied external stress. In covalent nanostructures regarded in the main text, this really leads to the reproduction of hydrostatic compression conditions in simulations, whereas in delaminated structures nothing can prevent core atoms from going back to the state close to equilibrium (practically without mechanical stress). Therefore, $\langle k \rangle_0$ becomes numerically overestimated, does not provide any physical meaning, and this approach cannot be directly applied for layered structures.

However, as mechanical stiffness of onion-like structures is fully determined by the outer shell (for any reasonable strain), it is possible to remove the core and consider only the outer shell apart. This is equivalent to considering some fullerene-like structure, and obtained results and trends for its average bond stiffness would be practically identical to fullerenes which are already discussed in the main text.

As for partially graphitized structures like studied truncated octahedral nanodiamonds, this is the trickiest case for the evaluation of average bond stiffness because core and shell are neither clearly separated to regard them as onions, nor sufficiently bonded to consider them as a covalent system. The same problems as in the case of octahedral nanodiamonds with direct calculation of $\langle k \rangle_0$ arise here. $\langle k \rangle_0^{est}$ could be calculated, however, the obtained number will not be a correct characteristic of uniform hydrostatic compression. Thus, other approaches should be proposed for such structures.

S5.2. Defective structures

We found that in principle, $\langle k \rangle_0$ is capable of describing mechanical properties even for defective structures, albeit with some modifications. We performed additional DFT calculations for structures of diamond and graphene containing point defects. We considered 2×2×2 diamond supercell and 6×6×1 graphene supercell with 1 vacancy. We also considered the same 6×6×1 supercell of graphene with Stone-Wales (SW) defect (Figure S3). The calculations were done with the same computational setup as in the main text.

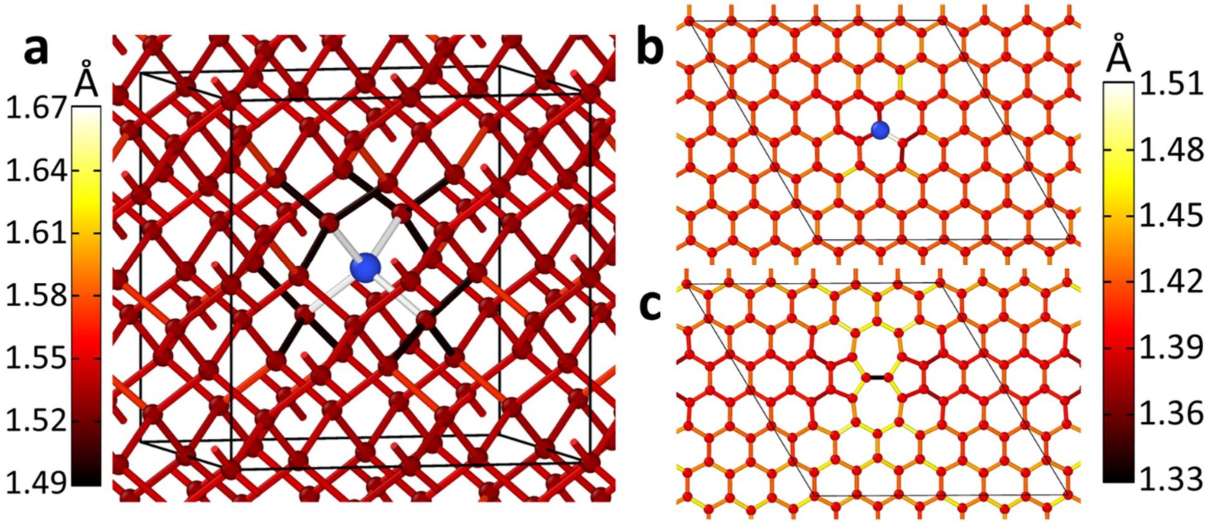

Figure S3. Relaxed atomic structures of (a) bulk diamond with vacancy and graphene with (b) vacancy and (c) Stone-Wales defect. Bonds are colored according to colormaps in the left (a) and in the right (b, c). Supercells used in calculations are shown with black lines. Blue atom in frames (a) and (b) is a dummy atom inserted at the place of vacancy. Dummy atom was introduced *post factum*, after the relaxation of structures with a vacancy.

After geometry optimization, these structures were uniformly strained to evaluate their mechanical stiffness. Using the corresponding equation of state, we calculated bulk (layer) modulus for defective diamond (graphene). The results for B_0 and γ_0 presented in Table S1 below witness that the presence of vacancy leads to a noticeable softening of both graphene and diamond, whereas SW defect only slightly lowers stiffness of graphene.

Further, we tried to apply the average bond stiffness approach to these structures. Our method succeeds without any modifications in the case of SW defect because this type of defect does not lead to breakage of covalent bonds and in fact, represents just a local distortion of the honeycomb lattice. As for the description of the structures with a vacancy, some modifications need to be done to the method.

Table S1. Mechanical stiffness of pristine and defective graphene and diamond. Values of B_0^{est} and γ_0^{est} were calculated from values of $\langle k \rangle_0$ using Eq. (3) and (4) from the main text.

	Graphene			Diamond	
	pristine	vacancy	SW defect	pristine	vacancy
Average bond stiffness $\langle k \rangle_0$ [N m^{-1}]	719	638.2	713.5	465.6	444.0
Directly calculated B_0 [GPa] and γ_0 [N m^{-1}]	207.6	181.8	202.2	434.6	412.2
B_0^{est} [GPa] and γ_0^{est} [N m^{-1}] estimated using $\langle k \rangle_0$	207.6	184.4	205.5	434.6	414.0

One can introduce a fictitious atom in the center of space between near-vacancy atoms (Figure S3) and then assign fictitious bonds between a dummy atom and near-vacancy atoms. Hence these fictitious bonds are introduced, average bond length changes (relatively to the same structure with a vacancy but without a dummy atom). Also, energies previously computed by DFT (without dummy atom), are renormalized, since the number of bonds changed. Here we emphasize that introduced additional atom and bonds do not contribute to the total energy, and thus these bonds effectively have zero stiffness, but finite length and number. This way, the interaction of unpaired electrons appearing in the vicinity of vacancy can be effectively taken into account in the framework of our simple model, and the values of $\langle k \rangle_0$ obtained after such a modification are in good agreement with directly calculated B_0 or γ_0 (see Table S1).

References

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